Activity Selection — Detailed Greedy Algorithm Notes

Topics: Normal explanation, mathematical formulation, greedy choice, exchange argument, optimal substructure, dry run, correctness, algorithm, complexity, and concept map.

1. Problem Statement

We are given n activities. Each activity has a start time S[i] and an end time E[i]. A person can perform only one activity at a time. The goal is to select the maximum number of activities such that no two selected activities overlap.

Example

```
start = {1, 3, 0, 5, 8, 5}
end   = {2, 4, 6, 7, 9, 9}
```

Activity	Start	End
A0	1	2
A1	3	4
A2	0	6
A3	5	7
A4	8	9
A5	5	9

One optimal answer is: **A0** → **A1** → **A3** → **A4**, so the answer is **4**.

2. What Does Non-Overlapping Mean?

If activity i is selected and another activity j is selected after it, then the next activity must start at or after the previous activity's end time.

Compatibility condition: S[j] ≥ E[i]

Example: A0 = 1→2 and A1 = 3→4. Since 3 ≥ 2, they do not overlap. But A0 = 1→2 and A2 = 0→6 cannot both be selected because 0 ≥ 2 is false.

3. Main Greedy Idea

Always select the activity that finishes earliest.

An activity that finishes earlier leaves more time for future activities. For example, if A = 1→4 and B = 2→3, choose B because 3 < 4.

4. Why Earliest Ending Time?

Suppose two possible activities are X=(Sx, Ex) and Y=(Sy, Ey), with Ex < Ey. After selecting X, the available time for future activities starts at Ex; after selecting Y it starts at Ey. Since Ex < Ey, X leaves more time available.

Greedy rule: choose the activity with minimum ending time.

5. Mathematical Formulation

Let the activities be A1, A2, …, An. Each activity has a pair (Si, Ei), where Si is its start time and Ei is its end time. We seek a subset Ai1, Ai2, …, Aik such that:

E[i1] ≤ S[i2], E[i2] ≤ S[i3], E[i3] ≤ S[i4], …

Objective: **maximize k**, i.e., maximize the number of selected compatible activities.

6. Greedy Choice

Let Ag be the activity having the smallest ending time. Then Eg = min(E1, E2, …, En). The greedy algorithm selects Ag first and then considers only activities compatible with it.

7. Why Is the Greedy Choice Correct?

Let G be the activity selected by the greedy algorithm and O be the first activity in some optimal solution. Because greedy chooses the earliest finishing activity:

E[G] ≤ **E[O]**

Therefore, G finishes no later than O. If we replace O with G in the optimal solution, every later activity that was compatible with O remains compatible with G. Hence the number of selected activities does not decrease.

Conclusion: There exists an optimal solution containing the greedy choice. Therefore, the greedy choice is safe.

8. Exchange Argument

Suppose an optimal solution begins O, A2, A3, … and greedy chooses G. Since E[G] ≤ E[O], replace O with G. The new solution G, A2, A3, … remains feasible and has the same number of activities. This proof technique is called the **Exchange Argument**.

9. Optimal Substructure

After selecting the greedy activity G = 1→2, only activities with S[i] ≥ 2 can be considered. The remaining task is another Activity Selection problem of the same type. Thus the problem has optimal substructure.

```
Original Problem
      ↓
Choose earliest finishing activity
      ↓
Remaining Problem
      ↓
Choose earliest finishing activity
      ↓
Repeat
```

10. Full Mathematical Dry Run

```
Start = [1, 3, 0, 5, 8, 5]
End   = [2, 4, 6, 7, 9, 9]

A0 = (1,2)
A1 = (3,4)
A2 = (0,6)
A3 = (5,7)
A4 = (8,9)
A5 = (5,9)
```

Step 1: Minimum ending time = min(2,4,6,7,9,9) = 2. Select A0=(1,2). Count=1.

Step 2: A1: 3 ≥ 2 → Select. lastEnd=4, count=2.

A2: 0 ≥ 4 is false → Skip.

A3: 5 ≥ 4 → Select. lastEnd=7, count=3.

A4: 8 ≥ 7 → Select. lastEnd=9, count=4.

A5: 5 ≥ 9 is false → Skip.

Final answer = 4. Selected activities: A0 → A1 → A3 → A4.

11. Timeline

```
A0: 1 ■■ 2
A1:       3 ■■ 4
A3:             5 ■■■■■ 7
A4:                       8 ■■ 9
```

The selected activities do not overlap.

12. Why Not Choose A2?

A2 = 0→6 is long. If we select A2 first, we can only select A4 = 8→9 afterward, giving only 2 activities. Greedy instead selects A0 → A1 → A3 → A4, giving 4.

13. Why Not Choose Earliest Start?

```
A0 = 0 → 100
A1 = 1 → 2
A2 = 3 → 4
A3 = 5 → 6
```

Choosing earliest start selects A0 and gives only 1 activity. Choosing earliest finish gives A1 → A2 → A3, giving 3. Therefore earliest start is not the correct greedy criterion.

14. Why Not Choose Shortest Activity?

```
A0 = 1 → 2
A1 = 2 → 100
A2 = 3 → 4
A3 = 4 → 5
```

Shortest duration is not the proven criterion. The correct criterion is min(E[i]), not min(E[i]−S[i]).

15. Algorithm

```
If activities are already sorted by end time:

1. Select the first activity.
```

```
2. Set lastEnd = its end time.
3. Traverse the remaining activities.
4. If start[i] >= lastEnd:
       Select activity
       count++
       lastEnd = end[i]
5. Return count.
```

16. Pseudocode

```
count = 1
lastEnd = end[0]

for i = 1 to n-1:
    if start[i] >= lastEnd:
        count++
        lastEnd = end[i]

return count
```

17. If Activities Are Not Sorted

First sort activities by increasing end time, then apply the greedy algorithm.

Time Complexity: Sorting O(n log n) + traversal O(n) = O(n log n). If already sorted, traversal is O(n).

Space Complexity: O(1) extra space if using the arrays directly; O(n) if an additional activity-index array is used.

18. Most Important Mathematical Idea

Let G be the greedy choice and O be the first activity of an optimal solution. The key inequality is:

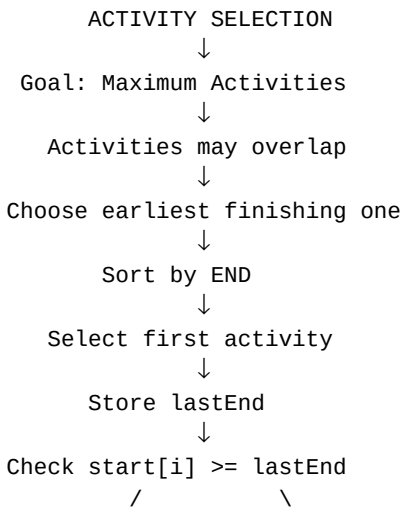
E[G] ≤ **E[O]**

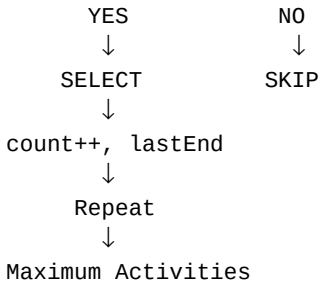
Because G finishes no later than O, replacing O with G cannot make the remaining activities impossible. Hence the greedy choice is safe. Repeat the same reasoning for the remaining problem.

19. Greedy Correctness — Exam/Interview Answer

We select the activity with the earliest finishing time because it leaves the maximum amount of time for the remaining activities. Suppose an optimal solution selects another activity first. Since the greedy activity finishes no later than that activity, we can replace the optimal solution's first activity with the greedy activity without decreasing the number of activities selected. Therefore, the greedy choice is safe. We then apply the same strategy to the remaining activities.

20. Final Concept Map





21. Quick Revision

One sentence: In Activity Selection, always choose the activity with the smallest end time because it leaves the maximum possible time for the remaining activities.

Greedy choice: min(E[i]) **Compatibility:** S[i] ≥ lastEnd **Objective:** maximize the number of selected activities.